

21 labels are enough: frozen foundation models for stem cell colony QC

From-scratch CNN 0.65 $\rightarrow$ frozen-feature ensemble 0.86. McNemar $p < 10^{-6}$, $n = 269$.

Ashkan Motamedifar¹ · Narges Hassanaghahi¹

¹University of Strasbourg, France

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1 Introduction

Human pluripotent stem cell (hPSC) colonies must be screened daily for quality. Manual scoring is slow, subjective, and a bottleneck for high-throughput culture.

A from-scratch CNN baseline (Mamaeva *et al.*, 2022) reached ≈ 0.89 accuracy on this dataset using heavy preprocessing and augmentation.

Research question. Can frozen features from *pretrained* or *self-supervised* vision foundation models match — or beat — a CNN trained end-to-end, on only 269 phase-contrast micrographs?

2 Method Overview

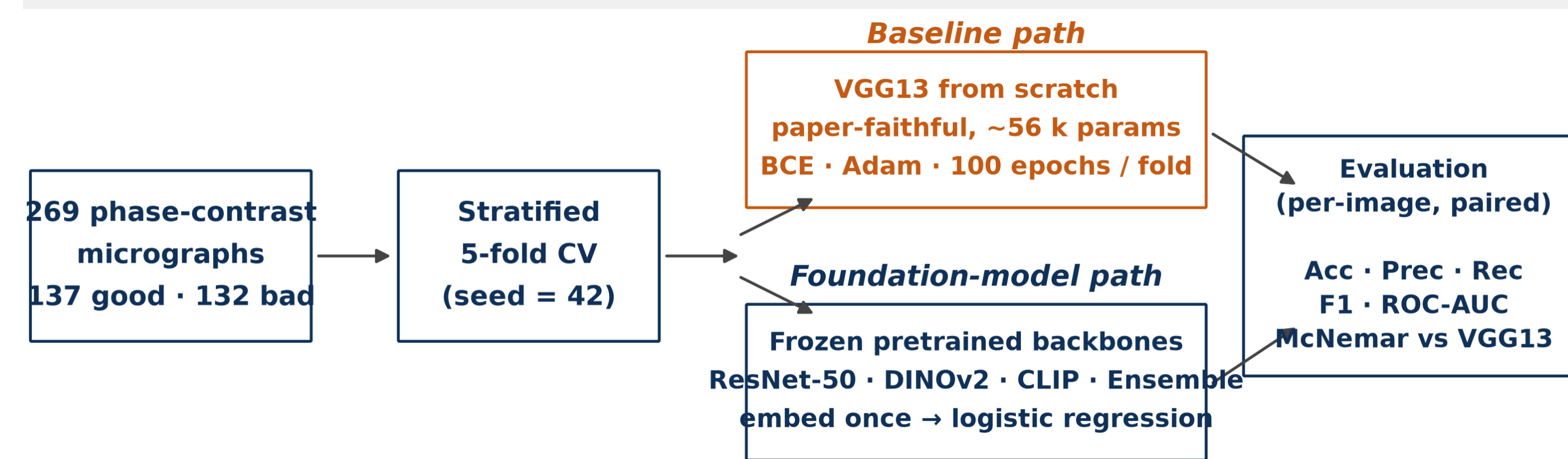


Figure 1: From dataset to evaluation — one shared pipeline for the from-scratch baseline and the foundation-model probes.

Four paradigms compared:

- **VGG13** — from-scratch CNN baseline.
- **ResNet-50** — ImageNet pretrained, linear probe.
- **DINOv2** — self-supervised, linear probe.
- **CLIP** — vision-language, linear probe + zero-shot.

Ensemble. Concatenate ResNet-50 + DINOv2 + CLIP embeddings, then a single logistic regression.

Training & evaluation.

- Input 224×224 , ImageNet normalisation.
- VGG13: BCE + Adam, 100 epochs / fold.
- Probes: logistic regression (L2 default).
- Paired McNemar tests vs. VGG13.



Code & data

3 Materials

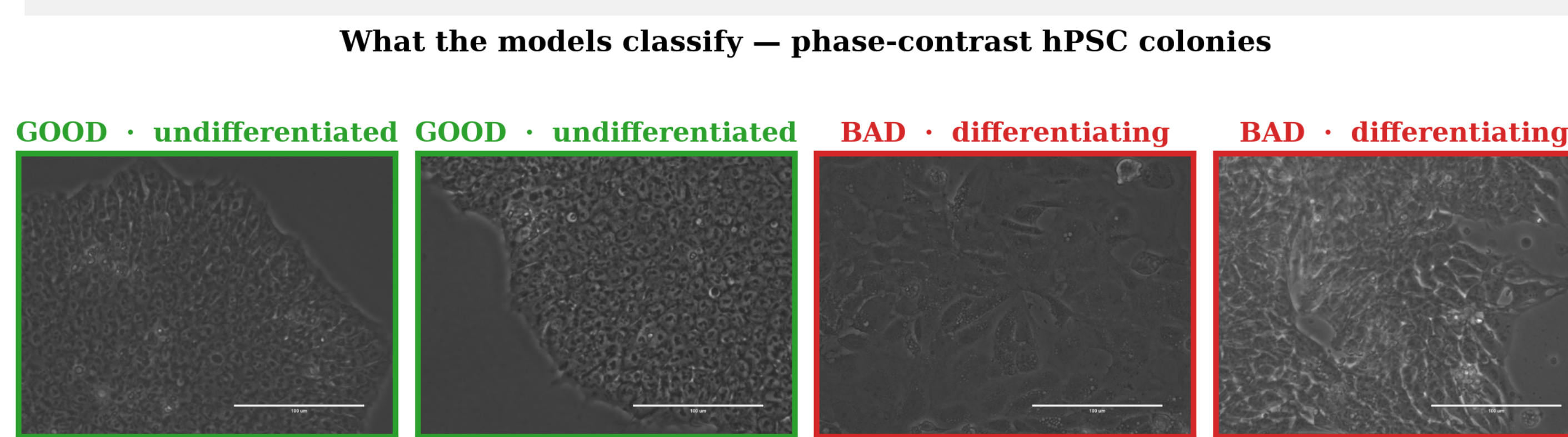


Figure 2: Example colonies. *Good*: compact borders, uniform morphology. *Bad*: rough edges, heterogeneous texture, or early differentiation.

Dataset.

- 269 phase-contrast micrographs (Zenodo 7316404).
- 137 *good* / 132 *bad* (near-balanced).
- 5-fold stratified splits, shared across all models.
- No augmentation; only resize & normalise.

5 Conclusion

- Frozen foundation features lift accuracy **0.65** $\rightarrow$ **0.86** (+21 pts) over from-scratch VGG13.
- 3-encoder ensemble: **0.93 ROC-AUC**, balanced errors.
- Probes beat the baseline with only ~ 21 labelled colonies per fold (Fig. 6).
- CLIP zero-shot $\approx$ chance: transfer from the encoder, not language priors.

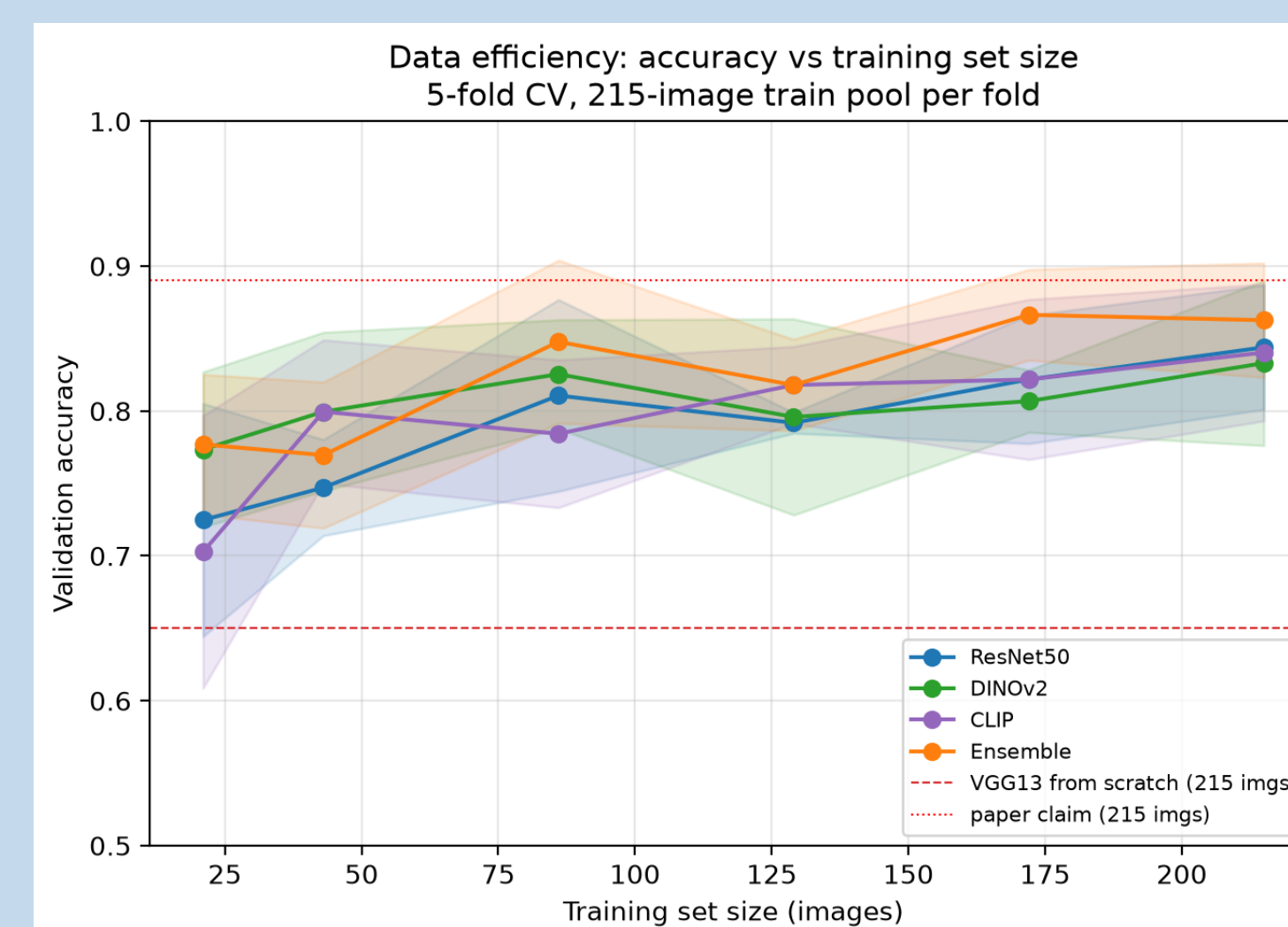


Figure 6: Label efficiency vs. training-set size.

Take-away. For small biomedical image sets, freeze a strong encoder and train a linear probe — skip the from-scratch CNN.

4 Results and Discussion

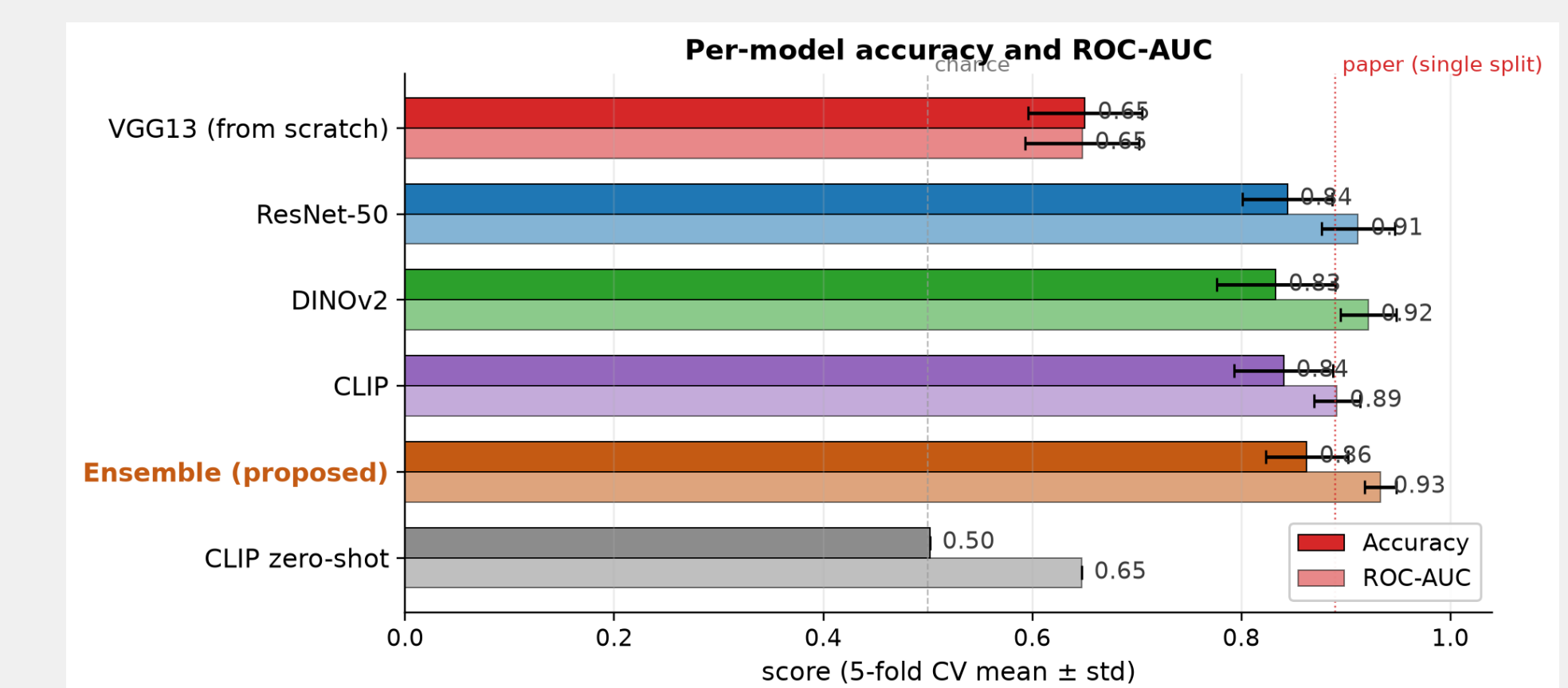


Figure 3: Accuracy and ROC-AUC across 5 folds. Ensemble reaches 0.86 accuracy / 0.93 AUC.

	VGG13 (from scratch)	acc = 0.65	Ensemble (proposed)	acc = 0.86	CLIP zero-shot	acc = 0.50
bad	44	88	111	21	0	132
good	6	131	16	121	2	135
	bad	good	bad	good	bad	good
	predicted		predicted		predicted	

Figure 4: VGG13 over-calls *bad* as *good*; CLIP zero-shot collapses; ensemble stays balanced.

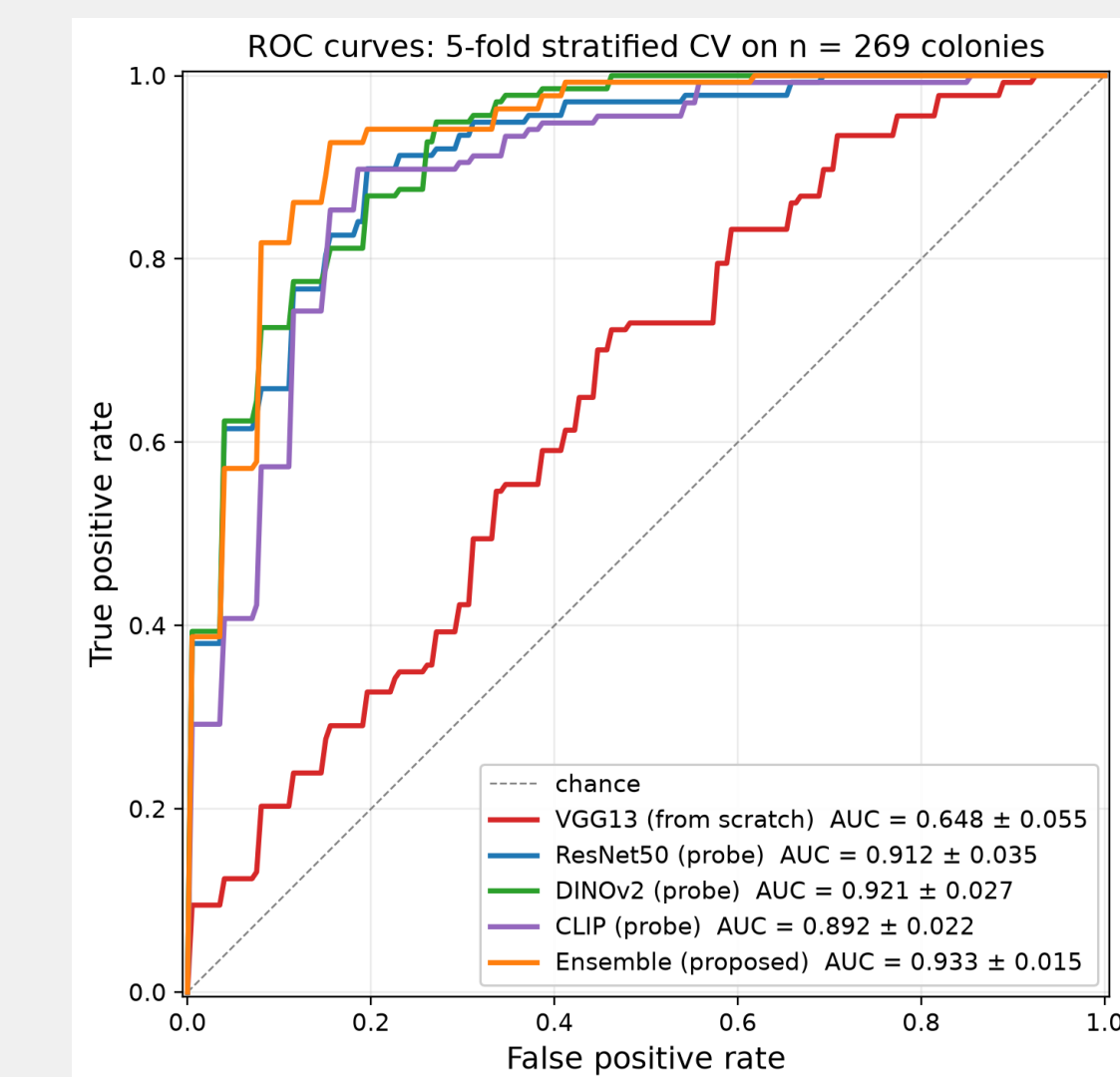


Figure 5: ROC curves. All pretrained probes lift above the VGG13 diagonal; ensemble AUC ≈ 0.93 .

References

1. Mamaeva *et al.* Quality control of hPSC colonies via deep learning. *Int. J. Mol. Sci.* 24(1):140, 2022.
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- Dataset: doi.org/10.5281/zenodo.7316404 · Code: github.com/ashkan-motamedifar/stem-cell-colony-qc